

Airline Delay Analysis

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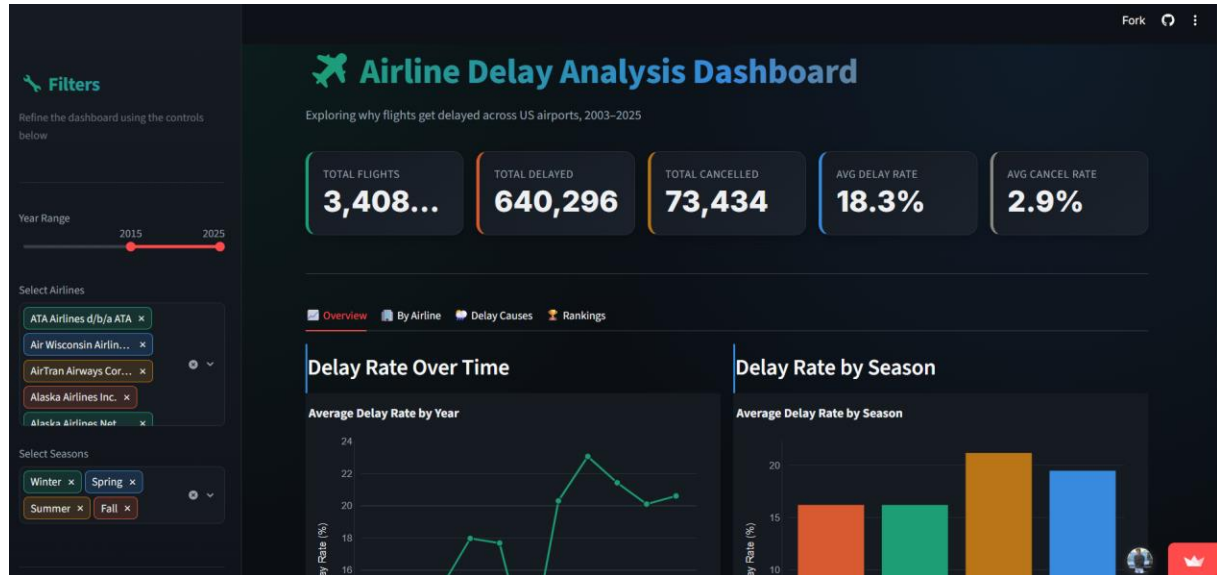
Subject: Data Visualization

Prof: Dina Deifallah

Snapshots and Insights:

Airline Delay Analysis Dashboard

Exploring why flights get delayed across US airports between 2003-2025

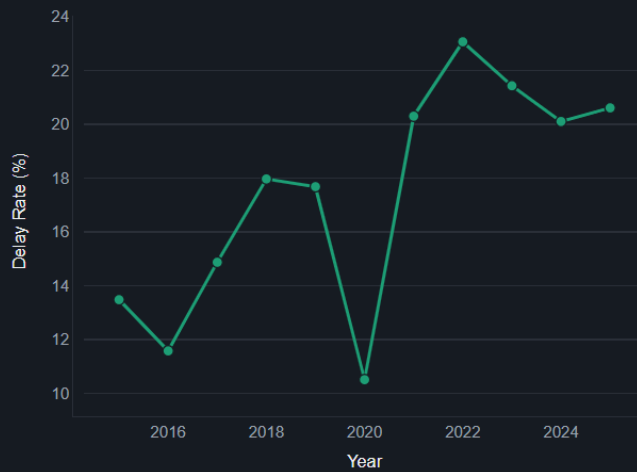


Delay Rate Over Time

Displays how the overall delay rate changes across different time periods.

Delay Rate Over Time

Average Delay Rate by Year



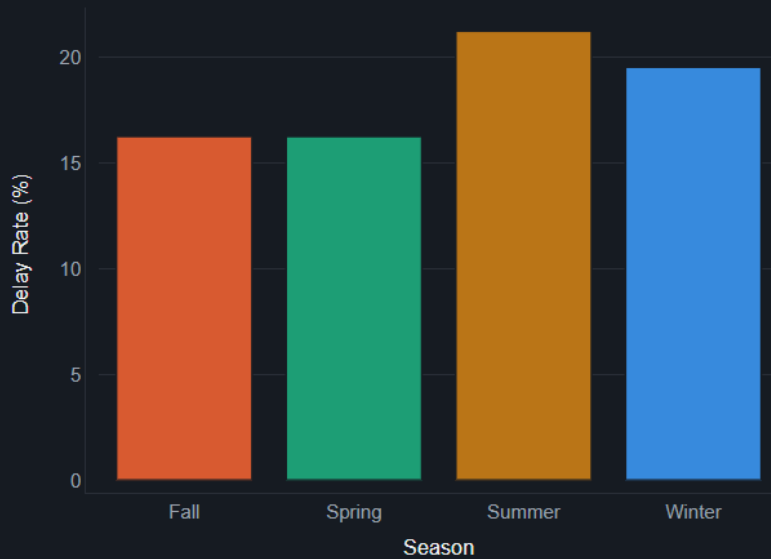
- The average delay rate shows noticeable fluctuations over time, indicating that flight punctuality is not consistent from year to year.
- The delay rate declines to its lowest point around 2020, suggesting improved on-time performance during that period.

Delay Rate by Season

Shows how delay causes vary across different seasons.

Delay Rate by Season

Average Delay Rate by Season



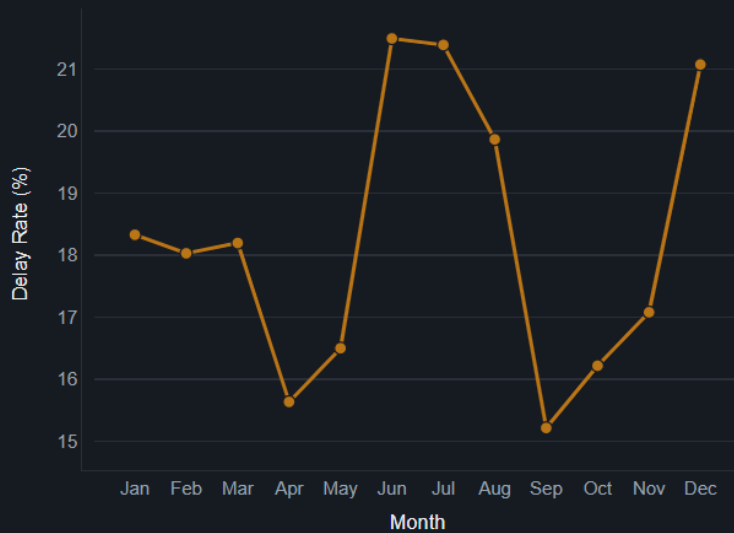
- Weather-related delays are generally more common during adverse seasonal conditions, while other causes remain relatively stable.
- Summer records the highest average delay rate, indicating that flights experience more delays during this season.
- The chart highlights a clear seasonal variation in flight delays, with delays generally increasing during peak travel or adverse weather seasons.

Monthly Delay Trend

This chart illustrates the average flight delay rate for each month of the year for the selected airlines and years

Monthly Delay Trend ↔

Average Delay Rate by Month



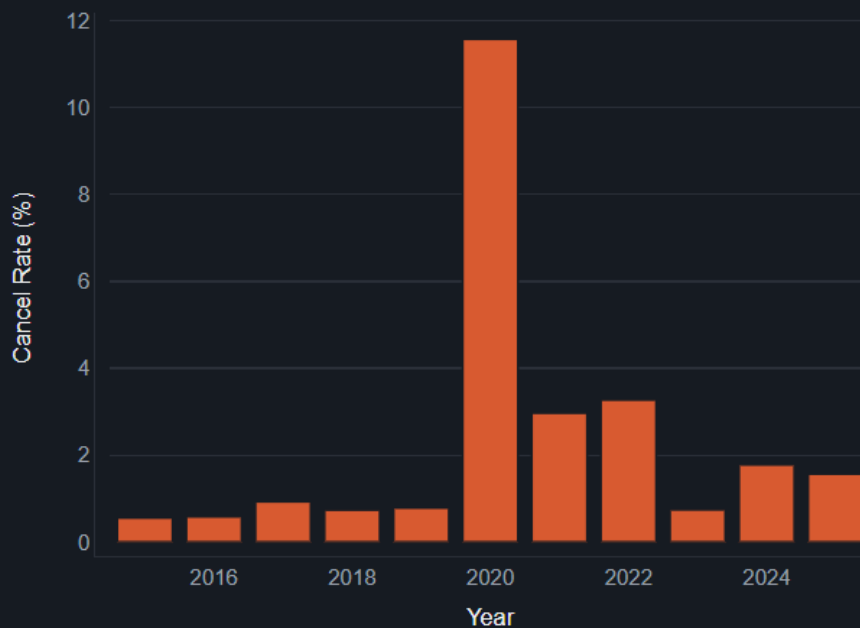
- June and July record the highest average delay rates, indicating increased flight delays during the peak summer travel season.
- The delay trend fluctuates throughout the year, showing that flight delays are influenced by seasonal demand and operational factors.

Cancellation Rate over time

Displays the average flight cancellation rate across different years

Cancellation Rate Over Time

Average Cancellation Rate by Year



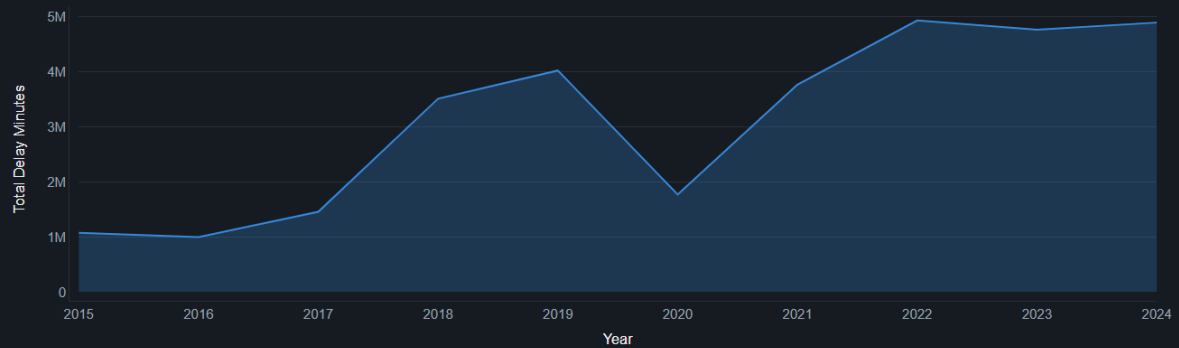
- The cancellation rate remains relatively low and stable for most years.
- The year-over-year comparison helps identify periods with unusually high flight cancellations

Total Delay Minutes over time

Displays the total delay minutes recorded across different years.

Total Delay Minutes Over Time

Total Delay Minutes by Year



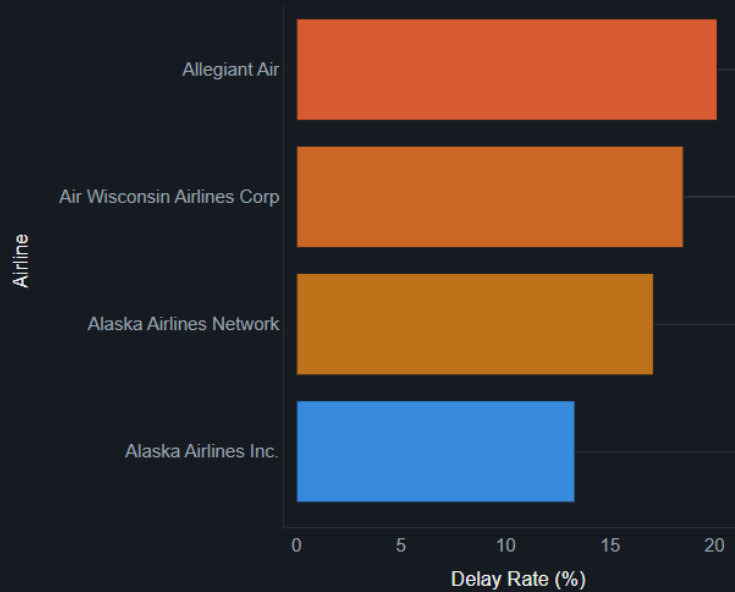
- Total delay minutes show an overall increasing trend over the years, indicating a rise in cumulative flight delays.
- The trend helps compare overall delay performance over time.

Airline-wise Delay Rate Analysis

Displays the percentage of delayed flights for each airline.

Delay Rate by Airline

Average Delay Rate by Airline



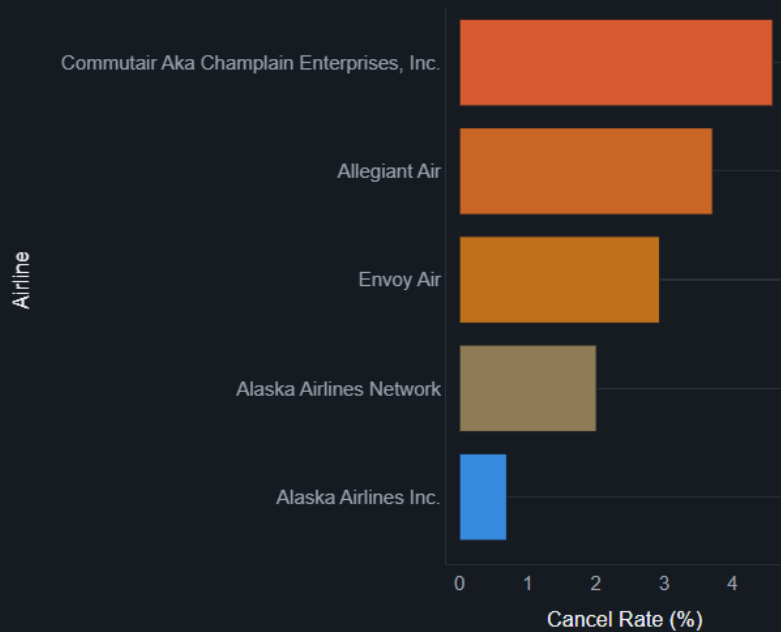
- Allegiant Air has the highest average delay rate.
- Alaska Airlines Inc. has the lowest delay rate among the airlines shown.
- Delay rates vary across airlines, highlighting differences in operational performance.

Airline-wise Cancellation Rate Analysis

Shows the cancellation percentage across different airlines.

Cancellation Rate by Airline

Average Cancellation Rate by Airline



- Commutair Aka Champlain Enterprises, Inc. Records the highest cancellation rate, indicating it experiences the most flight cancellations among the airlines displayed.
- Alaska Airlines Inc. Has the lowest cancellation rate among the airlines shown, reflecting more reliable flight operations.
- The chart highlights a clear variation in cancellation rates across airlines, indicating differences in operational efficiency and weather-related disruptions.

Delay minutes by cause

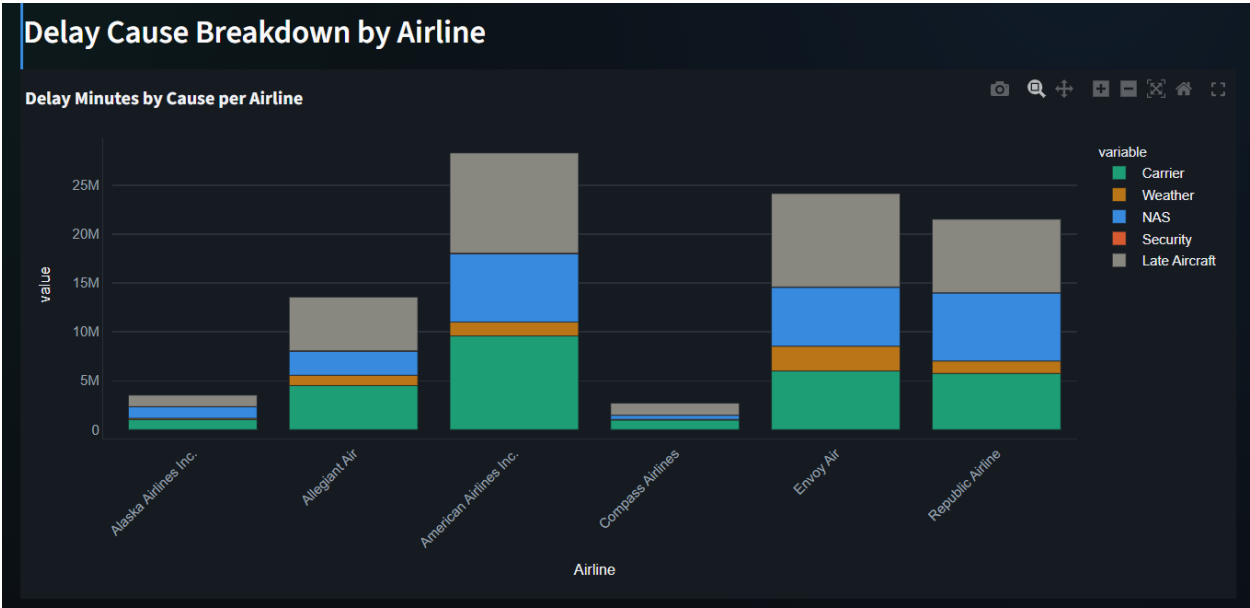
This chart shows the percentage contribution of different causes to the total flight delay minutes.



- Late Aircraft is the largest contributor to total delay minutes, accounting for approximately 37.2% of all delays.
- The majority of flight delays are caused by Late Aircraft, Carrier, and NAS-related issues, while Weather and Security contribute comparatively less. Focusing on operational efficiency, aircraft scheduling, and air traffic management can significantly reduce overall flight delays.

Delay cause breakdown by Airline

This chart displays the total delay minutes by airline, segmented by different delay causes to highlight the major contributors to flight delays for each airline.



- American Airlines recorded the highest total delay minutes, mainly driven by Late Aircraft and Carrier delays, indicating the greatest operational impact among the selected airlines.
- Security-related delays are minimal for all airlines, showing that operational and air traffic factors have a much greater influence on total delays.

Best & worst Airlines

🏆 Best Airlines (Lowest Delay Rate)			⚠️ Worst Airlines (Highest Delay Rate)		
Airline	Avg Delay Rate (%)	Total Flights	Airline	Avg Delay Rate (%)	Total Flights
Alaska Airlines Inc.	13.32	534869	Allegiant Air	20.14	765522
Republic Airline	14.63	2112629	American Airlines Inc.	19.63	2536827
Compass Airlines	16.81	206764	Envoy Air	17.32	2140409
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American Airlines Inc.	19.63	2536827	Republic Airline	14.63	2112629
Allegiant Air	20.14	765522	Alaska Airlines Inc.	13.32	534869

- The comparison highlights noticeable differences in delay rates across airlines, reflecting variations in operational efficiency, scheduling, and service reliability.
- Airlines with lower delay rates demonstrate better schedule adherence, while airlines with higher delay rates may benefit from improvements in operational planning and resource management.

GitHub Repository:

[GitHub - hetmodi2004/airline_delay_analysis · GitHub](https://github.com/hetmodi2004/airline_delay_analysis)

Live Dashboard:

[Airline Delay Dashboard · Streamlit](#)

Overall Conclusion

The Airline Delay Analysis Dashboard provides an interactive analysis of airline and airport performance using historical flight delay data. The visualizations identify trends in delay rates, seasonal patterns, cancellation behavior, major delay causes, and the best- and worst-performing airlines and airports. These insights can help airlines improve operational efficiency and assist passengers in making informed travel decisions. The interactive filters further enable users to explore the data dynamically based on airlines, years, and seasons.